

Shivam Agarwal

Email: shivamagarwal23012006@gmail.com

Phone: +91-9557284780

GitHub: [Shivam-Agarwal-2006](#)

LinkedIn: [shivam-agarwal](#)

Education

- **Indian Institute of Technology Roorkee** Roorkee, India
Bachelors and Masters in Mathematics and Computing. CGPA: 9.59 Aug 2024 - May 2029 (Expected)
- **Doon Public School** Roorkee, India
Class XII, CBSE. Percentage: 95.2% April 2023 - March 2024
- **Achievers Home Public School** Haridwar, India
Class X, CBSE. Percentage: 94.2% April 2021 - March 2022

Projects

- **Encryptify — Encryption and Decryption Web Application** [Source Code](#)
React, Tailwind CSS, Node.js, Express, MongoDB June 2025
 - Designed and developed a web-based encryption/decryption tool supporting secure text transformation using classical cryptographic techniques.
 - Implemented client-side and server-side logic to process encryption requests with focus on correctness and performance.
 - Focused on clean UI and usability, providing instant encryption/decryption results.
 - Deployed the application for public access with proper environment configuration.
- **AutoJudge — Programming Problem Difficulty Prediction** [Source Code](#)
Python, scikit-learn, Pandas, NumPy December 2025
 - Built a machine learning model to predict programming problem difficulty based on problem features.
 - Performed data cleaning, feature engineering, and exploratory data analysis on 4,000 problem samples.
 - Implemented classification models (Logistic Regression) and regression models (Random Forest Regressor) to evaluate difficulty prediction.
 - Evaluated performance using accuracy, precision, recall, F1-score, MAE, and RMSE metrics.
 - Finalized model tuning and validation for improved prediction performance.

Skills

- **Languages:** C/C++, HTML, CSS, JavaScript
- **Tools/Frameworks:** React.js, Node.js, Express.js, Tailwind CSS, MongoDB, Postman, PostgreSQL, Git/GitHub
- **Soft Skills:** Leadership, Team Working, Oratory, Initiative Skills
- **Professional Skills:** Object Oriented Programming(OOPs), Database and Management Systems(DBMS)
- **Key Courses:** Data Structures and Algorithms, Data Science, Statistics, Discrete Mathematics, Programming with C++, Database and Management Systems, Operation Research

Achievements

- Secured an All India Rank **625** and an All India Rank **1933** in JEE Main and Advanced 2024 respectively.
- State Topper (**Rank 1**) in JEE Main 2024 – Uttarakhand
- Solved over **500** problems on Data Structures and algorithms across various coding platforms.
- Attained highest rating of **1550** and the title of **Specialist** on **Codeforces**. Profile: [shivam2301](#)